

SINES MONTES CHAOS, Portugal

WMO: 085410

Lat: 37.9544N

Lon: 8.8383W

Elev: 324

StdP: 14.52

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.6	42.8	25.2	19.5	45.9	30.4	24.8	47.4	27.1	53.2	24.0	53.1	8.5	10	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.0	84.7	65.8	81.3	65.5	78.4	65.2	69.3	77.2	68.3	75.4	67.4	74.1	9.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.3	101.7	70.7	66.2	97.8	69.9	65.2	94.4	69.3	33.6	77.5	32.7	75.8	32.0	74.1	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	20.6	18.4	DB	35.0	93.8	2.9	5.1	32.9	97.5	31.2	100.4	29.6	103.3	27.4	107.0
			WB	31.5	71.8	3.0	2.8	29.3	73.7	27.5	75.4	25.9	76.9	23.7	78.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.2	53.0	53.3	56.3	58.6	62.6	66.7	68.0	69.5	68.1	65.1	58.0	54.4	(6)
(7)		DBStd	7.10	3.88	3.76	3.87	3.90	4.46	3.93	3.28	3.59	3.07	4.08	4.41	3.95	(7)
(8)		HDD50	45	18	13	3	0	0	0	0	0	0	0	2	9	(8)
(9)		HDD65	1887	372	327	271	197	104	21	3	1	5	46	212	328	(9)
(10)		CDD50	4126	111	106	197	258	390	501	559	604	543	469	242	146	(10)
(11)		CDD65	492	0	0	1	5	28	72	97	139	97	50	3	0	(11)
(12)		CDH74	1999	0	0	10	29	167	326	398	625	291	151	2	0	(12)
(13)		CDH80	454	0	0	1	3	40	71	105	168	41	25	0	0	(13)
(14)	Wind	WSAvg	9.0	8.6	9.2	9.3	9.2	9.3	9.4	10.2	9.0	8.0	8.1	8.6	8.7	(14)
(15)	Precipitation	PrecAvg	20.1	2.7	2.3	1.9	1.9	1.3	0.3	0.0	0.1	0.9	3.0	3.3	3.2	(15)
(16)		PrecMax	31.3	10.6	6.6	5.6	5.4	3.1	1.4	0.2	0.8	3.4	5.8	10.7	8.7	(16)
(17)		PrecMin	12.5	0.0	0.2	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.3	0.2	0.2	(17)
(18)		PrecStd	4.6	2.0	1.6	1.3	1.2	1.0	0.4	0.0	0.2	1.0	1.7	2.6	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	66.2	74.9	77.5	85.6	87.1	88.7	90.7	85.4	83.4	72.6	66.8	(19)
(20)		0.4%	MCWB	56.4	58.4	58.4	62.0	64.7	67.3	67.3	66.7	66.0	63.4	62.2	58.4	(20)
(21)		2%	DB	61.6	63.2	69.2	72.6	78.2	81.1	82.4	84.2	80.6	77.9	68.4	63.5	(21)
(22)		2%	MCWB	57.0	56.2	57.7	60.7	63.0	65.8	65.8	66.4	65.1	63.6	61.6	58.7	(22)
(23)		5%	DB	60.2	61.1	65.2	68.8	73.9	77.4	77.9	80.3	77.4	74.3	66.4	62.0	(23)
(24)		5%	MCWB	56.4	55.9	57.0	59.9	62.6	65.0	66.0	66.4	65.4	63.8	60.9	58.4	(24)
(25)		10%	DB	59.0	59.4	62.5	65.4	70.2	74.4	74.8	76.7	74.7	71.5	64.6	60.7	(25)
(26)		10%	MCWB	55.8	55.6	56.4	58.8	61.6	64.8	65.5	66.4	65.4	63.3	59.9	57.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.6	60.5	62.0	64.7	67.5	70.0	70.2	71.3	70.5	68.8	66.2	62.0	(27)
(28)		0.4%	MCDWB	61.8	63.0	68.3	73.0	77.2	79.1	79.7	79.3	76.7	73.4	68.1	62.9	(28)
(29)		2%	WB	58.9	58.9	59.9	62.4	65.4	68.0	68.4	69.3	68.5	67.4	64.2	60.8	(29)
(30)		2%	MCDWB	60.2	60.7	64.5	69.5	74.2	77.3	76.9	77.7	73.9	71.2	66.7	62.2	(30)
(31)		5%	WB	57.7	57.7	58.7	60.9	63.8	66.5	67.1	68.2	67.5	66.2	62.6	59.4	(31)
(32)		5%	MCDWB	59.3	59.9	62.5	66.8	71.3	74.8	74.9	75.9	72.5	70.2	64.8	61.3	(32)
(33)		10%	WB	56.6	56.3	57.6	59.4	62.4	65.3	66.1	67.1	66.5	65.1	61.1	58.0	(33)
(34)		10%	MCDWB	58.5	58.9	61.1	64.3	68.4	72.7	73.2	74.3	71.7	69.1	63.6	60.3	(34)
(35)	Mean Daily Temperature Range		MDBR	10.6	11.7	12.5	12.6	13.9	14.2	14.3	15.0	13.4	12.4	11.3	11.0	(35)
(36)		5% DB	MCDBR	11.0	13.3	16.9	18.5	20.8	19.8	19.7	20.5	17.7	16.7	12.3	11.3	(36)
(37)			MCWBR	8.1	9.2	10.3	10.4	9.7	8.2	8.0	7.8	7.4	8.3	8.1	8.2	(37)
(38)			MCDBR	8.4	10.7	12.8	15.2	16.9	16.8	16.1	15.8	12.9	11.9	9.7	9.2	(38)
(39)		5% WB	MCWBR	7.3	8.4	8.9	9.6	8.7	7.7	7.5	7.2	6.4	7.1	7.7	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.343	0.374	0.374	0.382	0.386	0.371	0.386	0.383	0.364	0.340	0.330	(40)	
(41)		taud	2.508	2.467	2.385	2.396	2.400	2.405	2.435	2.404	2.429	2.494	2.520	2.508	(41)	
(42)		Ebn at Noon	271	281	281	287	285	283	286	280	275	270	264	261	(42)	
(43)		Edh at Noon	25	30	35	37	37	37	36	36	33	29	25	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	759	1067	1444	1831	2187	2389	2446	2202	1728	1202	839	685	(44)	
(45)		RadStd	92	123	139	134	147	126	81	73	104	96	96	75	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page